



REPORT ON SHOPWAVE E-COMMERCE

Prepared for:

University of London
ShopWave E-Commerce

UOL Student Number:

10243697

In partial fulfilment of the requirements for the module:

Business Analytics, Applied Modelling & Prediction [ST2187]

Submission Date: 1 March 2025

Word Count* : 1463/1500

*(excluding Table of Contents, Executive Summary, Footnotes, Labels, Figures & Equations)

EXECUTIVE SUMMARY

The rapid growth of **ShopWave** over recent years necessitates a comprehensive evaluation of its performance to enhance operational efficiency and sustain profitability. This report provides an in-depth analysis of **sales trends, profit margins, customer acquisition, logistics efficiency, and future projections** to aid senior management in strategic decision-making.

The **sales performance analysis** highlights seasonal fluctuations, with peak sales occurring in **March, June, September, and December**, driven by promotional events and festive periods.

The **profitability overview** reveals that while sales have increased, certain product categories—particularly **furniture**—face declining margins due to **aggressive discounting and high shipping costs**. In contrast, **technology and office supplies** remain the most profitable segments. A **logistics efficiency review** identifies that **standard-class shipping**, though widely used, contributes to **higher late delivery rates**, impacting customer satisfaction.

The **customer segmentation analysis** indicates that while consumer segments drive **over 50% of total revenue**, **new customer acquisition has slowed**, necessitating targeted marketing initiatives.

Forecasts suggest **sustained growth in sales and profitability**, provided that strategic adjustments—such as **refining discount structures, optimising shipping costs, and reallocating resources to high-performing products**—are implemented. To maintain competitive advantage, **ShopNova must focus on enhancing customer retention, improving operational efficiencies, and leveraging data-driven insights for expansion**.

Key recommendations include **revising discount strategies, strengthening high-margin product offerings, and optimising regional logistics** to drive sustainable growth. With these measures, **ShopWave** is well-positioned to achieve long-term success and market leadership.

TABLE OF CONTENTS

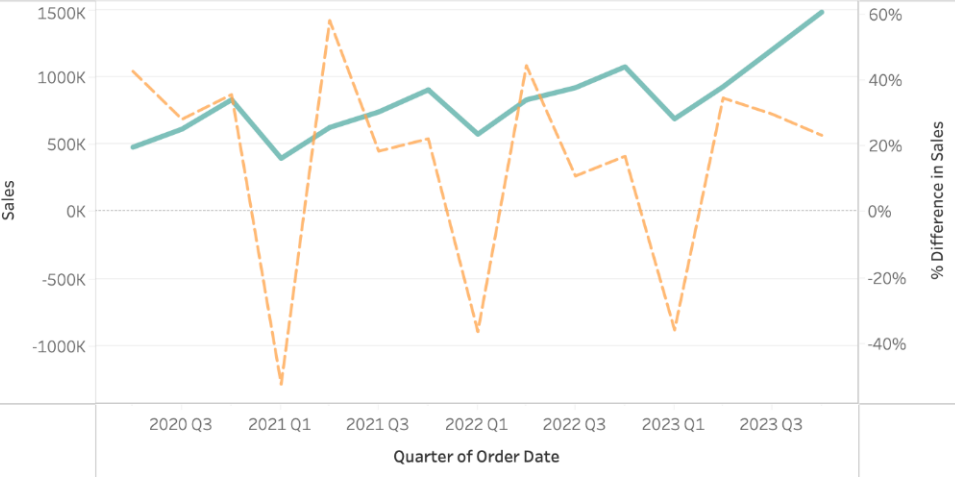
1. INTRODUCTION	4
2. SALES PERFORMANCE OVERVIEW	5
3. PROFIT ANALYSIS	8
4. LOGISTICS EFFICIENCY	11
5. CUSTOMER ACQUISITION & RETENTION	14
6. BUSINESS OUTLOOK & FORECAST	17
7. RECOMMENDATIONS	20
8. CONCLUSION.....	21

1. INTRODUCTION

ShopWave is a rapidly expanding online e-commerce platform, making waves in the retail space with its seamless and engaging shopping experience. As competition grows, senior management is committed to staying ahead of industry trends by leveraging data-driven insights to refine business strategies. This report evaluates key performance indicators (KPIs), including sales performance, profitability, customer retention, logistics efficiency, and market trends, to highlight areas for improvement and growth. With these insights, ShopWave aims to enhance customer satisfaction, optimise operations, and maintain long-term success.

2. SALES PERFORMANCE OVERVIEW

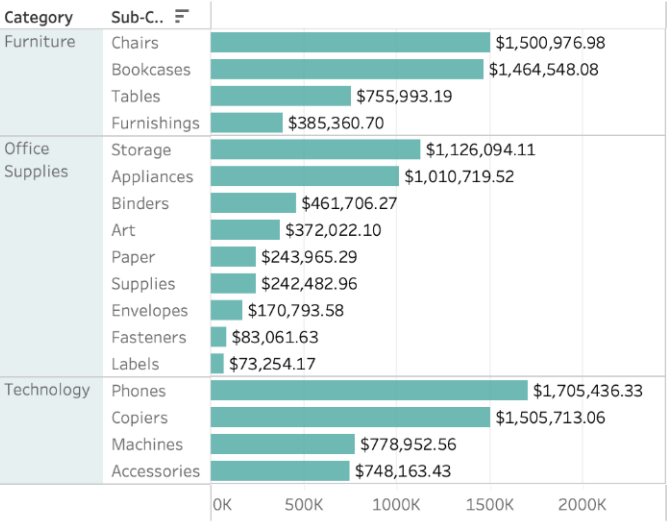
Quarterly Sales Trend
with growth % per quarter



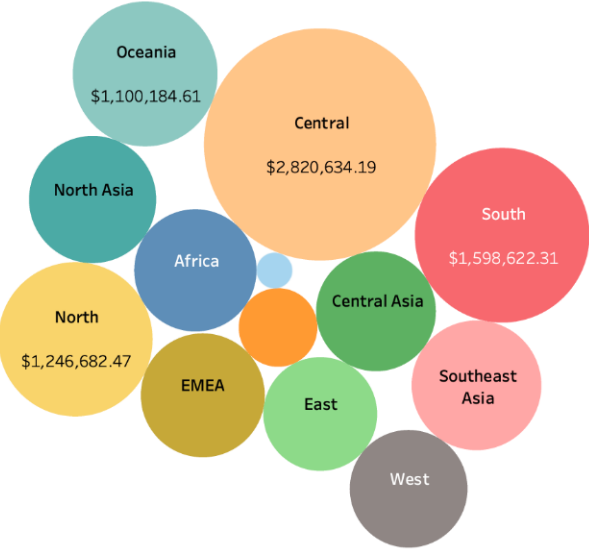
Annual Sales Overview



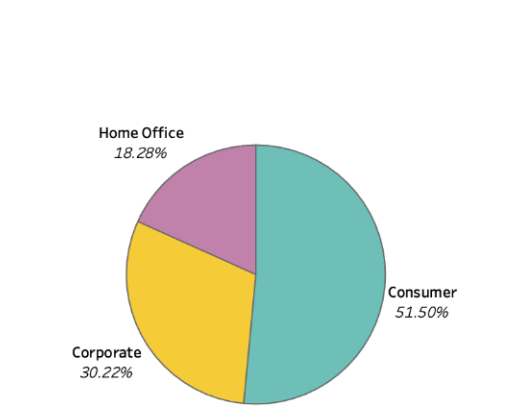
Sales per Category



Sales by Region



Sales by Segment



Dashboard 1: Sales Performance Overview Dashboard

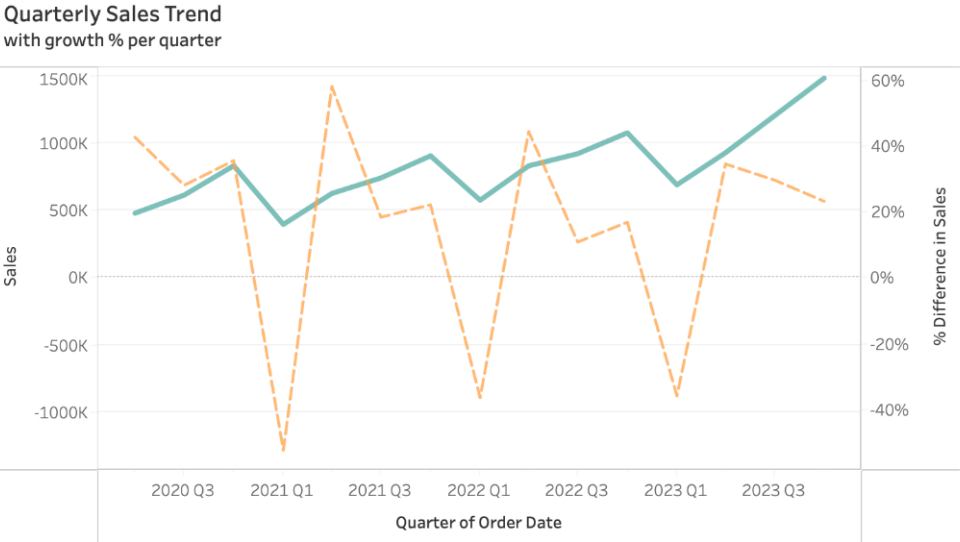


Figure 1.1

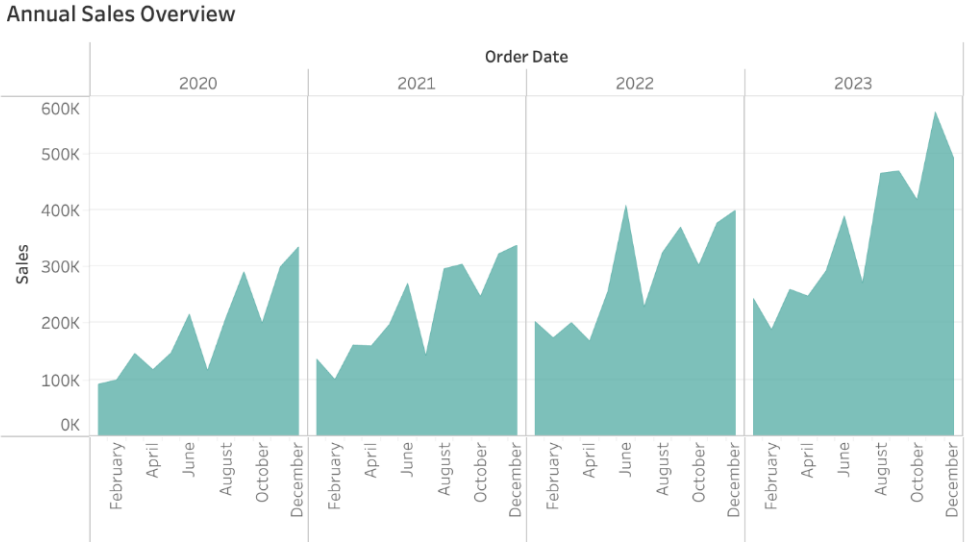


Figure 1.2

Revenue Trends

The revenue from 2020 to 2023 exhibits an upward trend (Figure 1.1), though quarterly fluctuations suggest seasonal patterns. The area under the graph in Figure 1.2, representing cumulative annual sales, shows a consistent increase over time, reinforcing revenue growth. Recurring peaks and troughs highlight the business's seasonality, with February, April, July, and October marking lower sales, while March, June, September, and December see heightened sales. The June increase is likely driven by mid-year promotions, while the December surge stems from year-end sales, festive seasons, and holiday spending.

Sales per Category

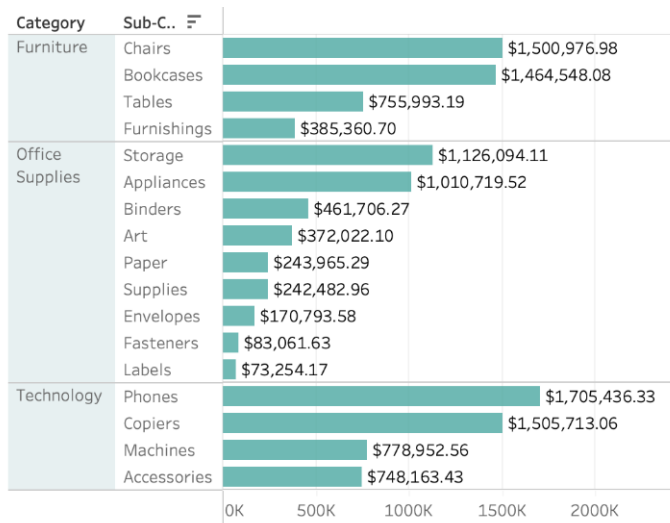


Figure 1.3

Sales by Region

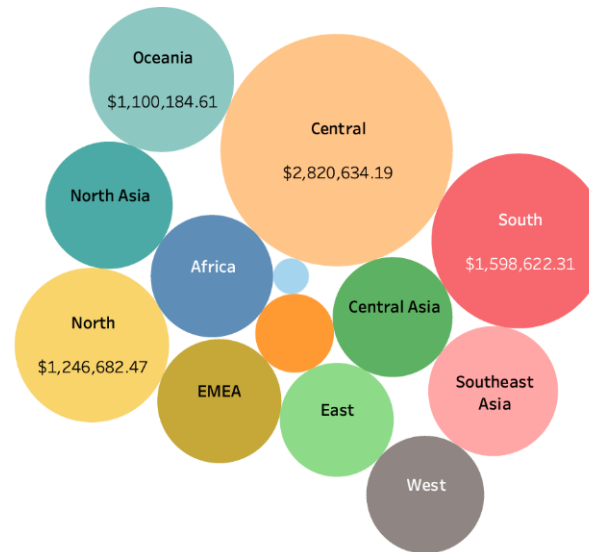


Figure 1.4

Sales by Segment

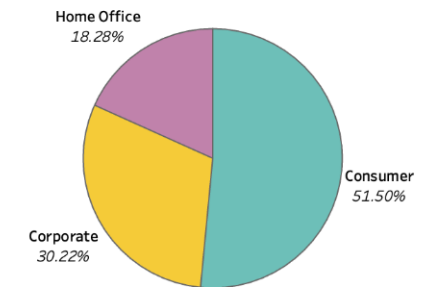


Figure 1.5

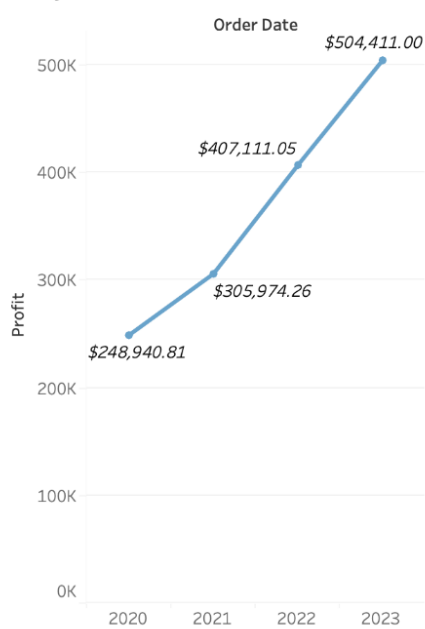
Category & Geographic Performance

Figure 1.3 highlights the top-selling products in each category: Chairs in Furniture, Storage in Office Supplies, and Phones in Technology. Identifying lower-selling products presents an opportunity to boost performance through targeted marketing, such as discounts or bundling with best-sellers. Market basket analysis can further optimise sales by identifying appealing product pairings.

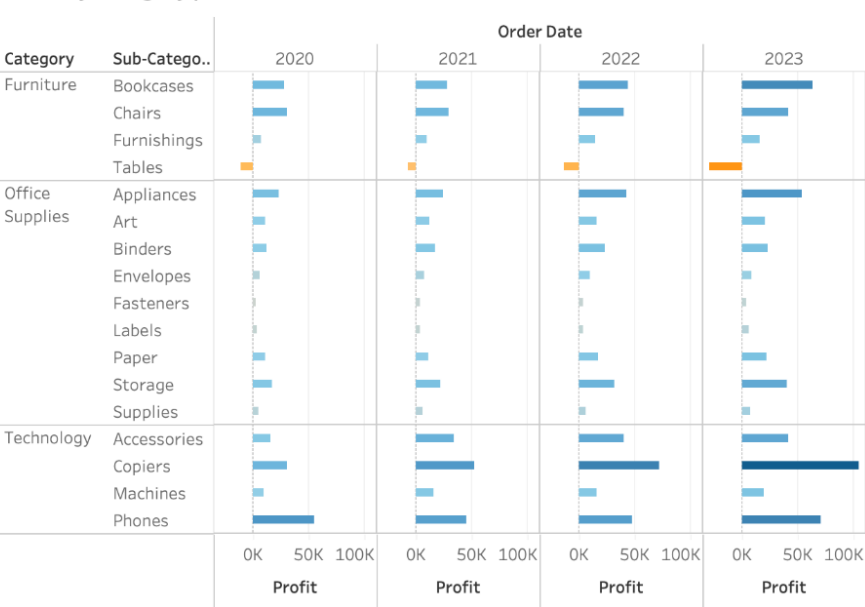
According to Figure 1.5, the Consumer segment is the largest contributor to ShopWave's sales at 51.50%, followed by Corporate at 30.22% and Home Office at 18.28%. Geographically, the Central region generates the highest sales at \$2.8 million, while North Asia records the lowest at \$847,000.

3. PROFIT ANALYSIS

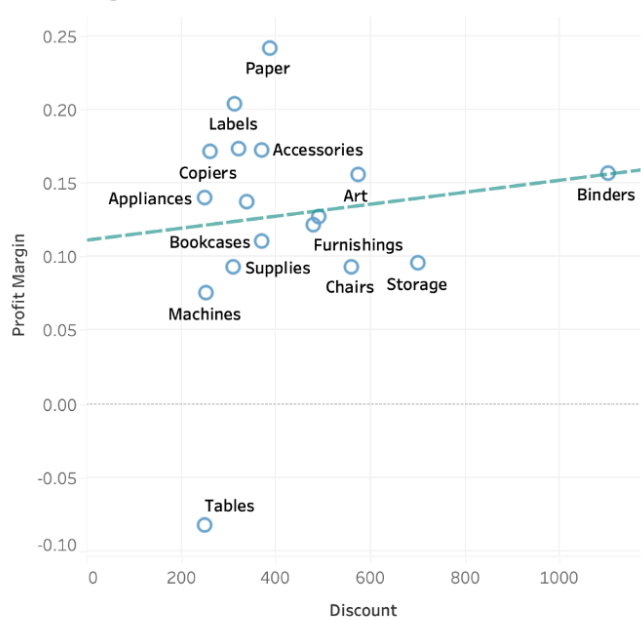
Yearly Profit



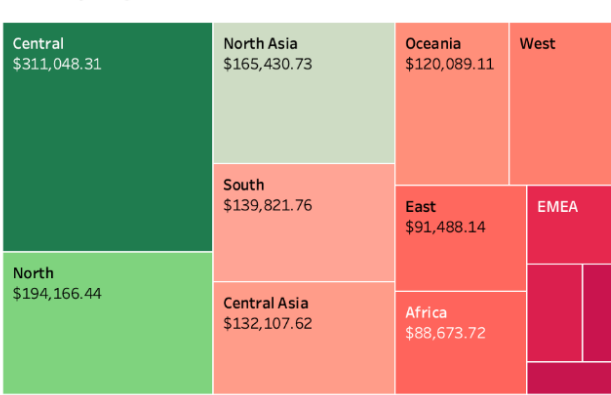
Profit by Category per Year



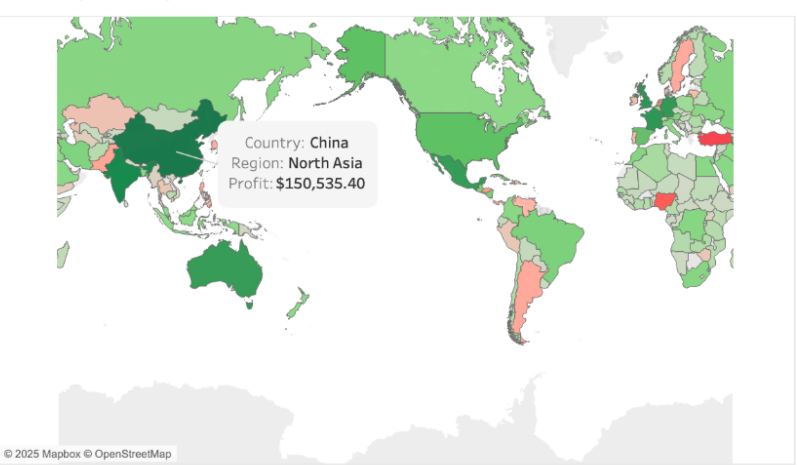
Profit Margin vs Discounts



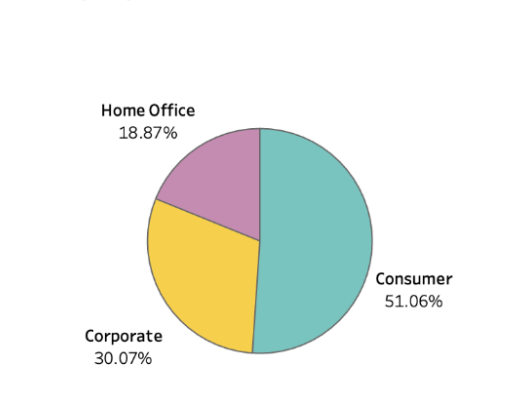
Profit by Region



Profit by Country



Profit by Segment



Profit \$17,817.39 \$311,048.31

© 2025 Mapbox © OpenStreetMap

Yearly Profit

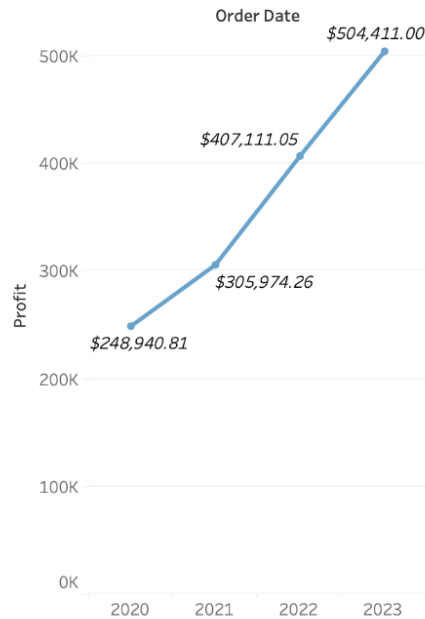


Figure 2.1

Profit by Category per Year



Figure 2.2

Profit Margin vs Discounts

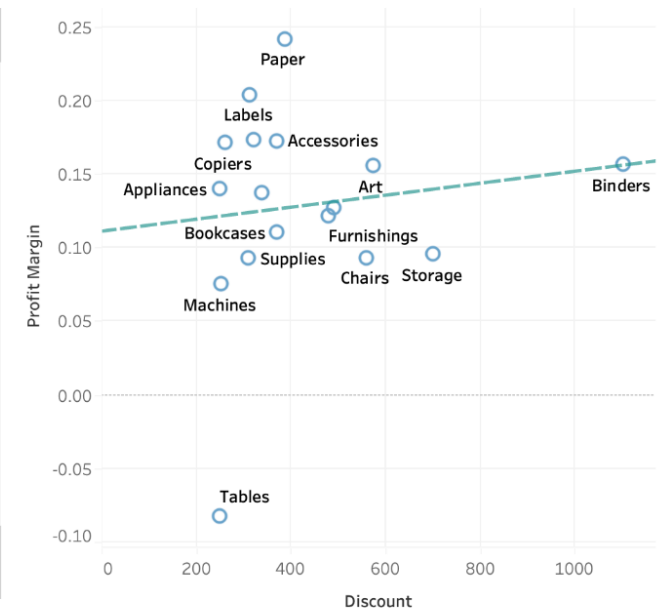


Figure 2.2

Profitability Trends and the Anomaly of Tables

Profitability has grown significantly from \$248.9K in 2020 to \$504.4K in 2023 (Figure 2.1), reflecting improved financial performance. Copiers are the most profitable product, while tables, highlighted in orange, consistently incur losses (Figure 2.2), likely due to high production costs, excessive discounting, or weak demand. Despite strong sales (shown previously in Figure 1.3), tables are sold at a loss, indicating that their costs exceed revenue, unlike other products where higher sales drive profitability. Figure 2.3 shows a general upward trend in profit margin with discounts, but tables deviate from this pattern, reinforcing their negative profitability and the need for a strategic review.

Profit by Region

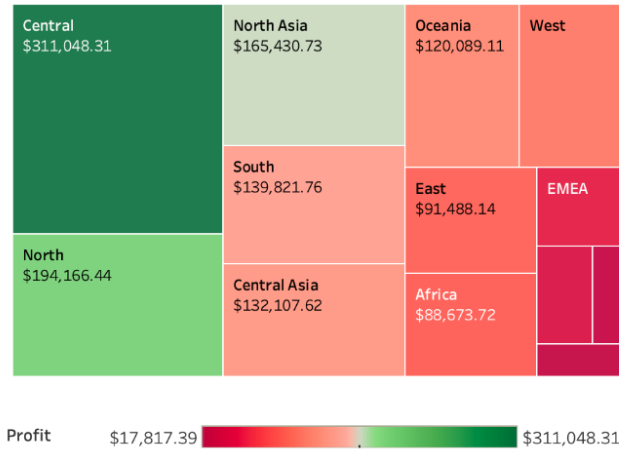


Figure 2.4

Profit by Country

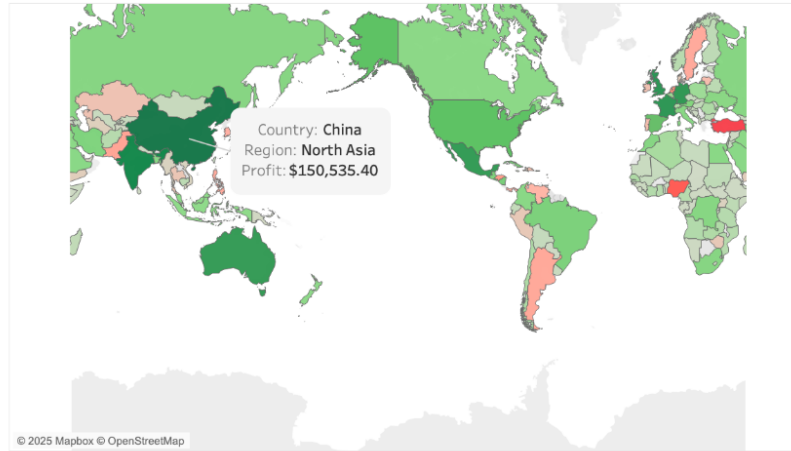


Figure 2.5

Profit by Segment

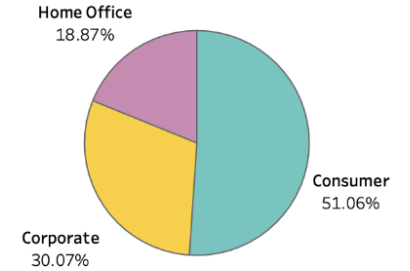


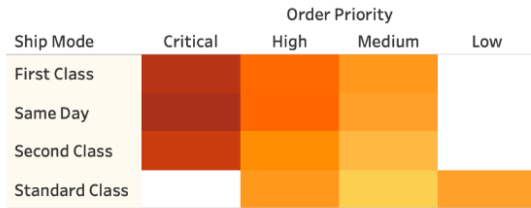
Figure 2.6

Regional and Customer Profitability Insights

The Central region records the highest profit, aligning directly with its strong sales performance (Figure 2.4). On a country level, China leads in profitability, generating \$150,535 in profit (Figure 2.5). In terms of customer segments, consumers contribute the most to overall profit, accounting for 51.06% of total profitability (Figure 2.6). These insights highlight the key geographic and demographic drivers of profitability, which can inform targeted business strategies.

4. LOGISTICS EFFICIENCY

Order Priority
vs shipping costs



Shipping Costs w.r.t Sales

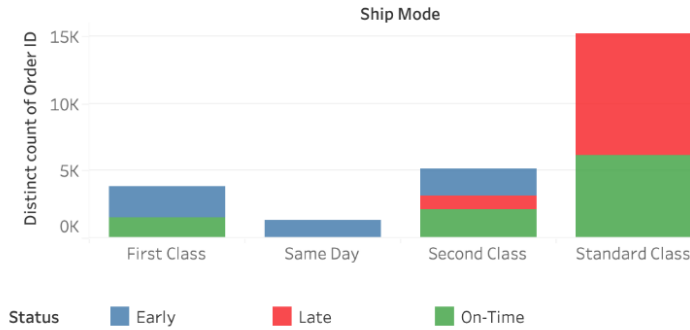
Ship Mode	Sales	Shipping Cost	Shipping Cost wrt Sales
First Class	1,827,897	307,651	16.83%
Second Class	2,560,286	313,401	12.24%
Standard Class	7,574,073	614,320	8.11%
Same Day	666,989	115,903	17.38%

Delivery Status

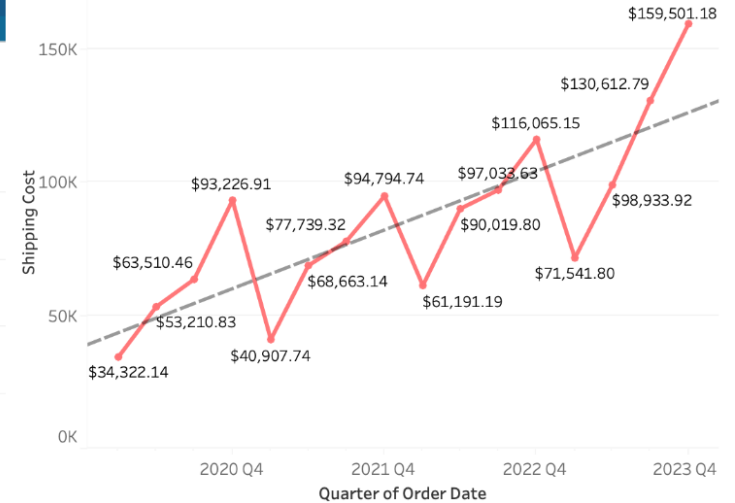


Delivery Status

Early → If the order is shipped within 2 days of the order being placed.
On-Time → If the order is shipped between 3-4 days after ordering.
Late → If the order is shipped more than 4 days after ordering.



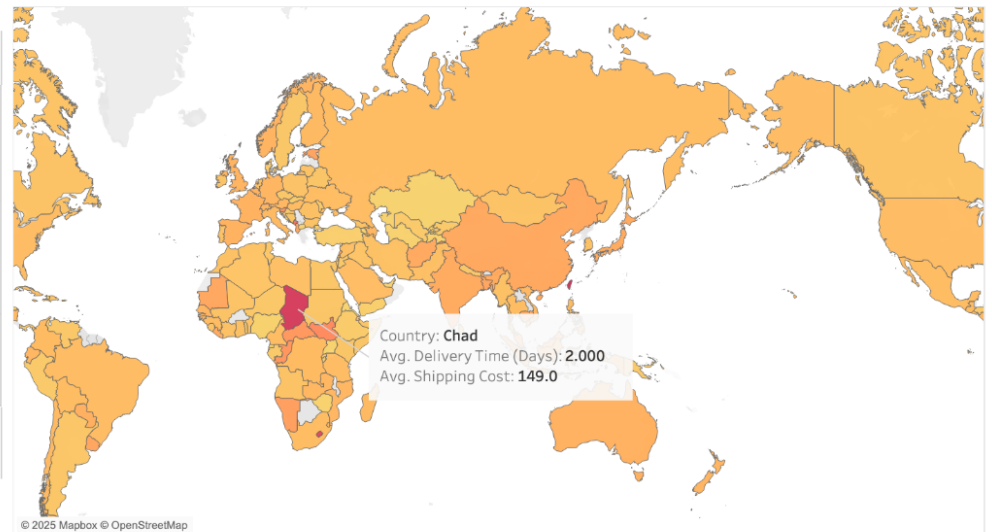
Delivery Cost
over the years

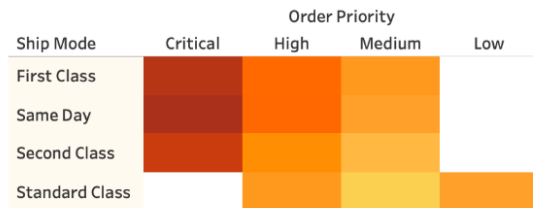


Actual Profits
post-Shipping cost deduction



Average Shipping Cost to Countries



Order Priority
vs shipping costs

Shipping Costs w.r.t Sales

Ship Mode	Sales	Shipping Cost	Shipping Cost wrt Sales
First Class	1,827,897	307,651	16.83%
Second Class	2,560,286	313,401	12.24%
Standard Class	7,574,073	614,320	8.11%
Same Day	666,989	115,903	17.38%

Figure 3.1

Delivery Status

Early	22.60%
Late	40.59%
On-Time	38.63%

Delivery Status

Early → If the order is shipped within 2 days of the order being placed.
 On-Time → If the order is shipped between 3-4 days after ordering.
 Late → If the order is shipped more than 4 days after ordering.

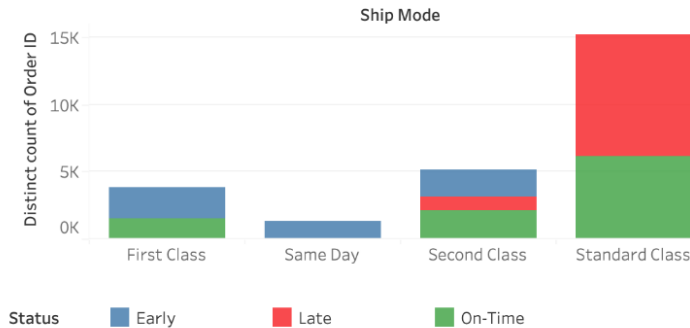


Figure 3.2

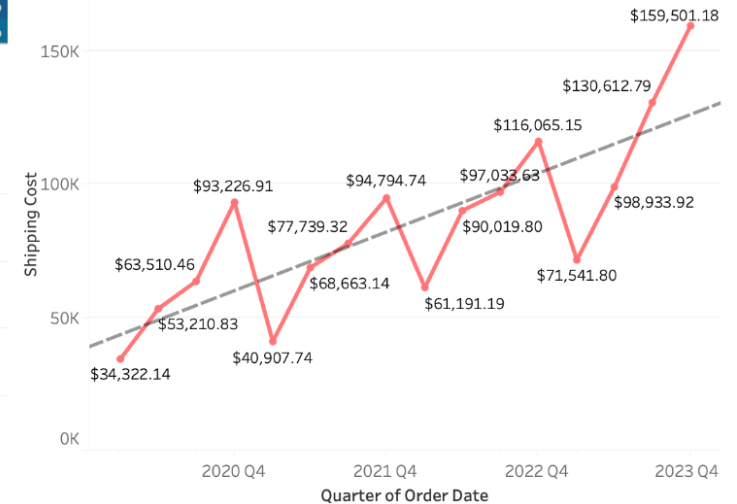
Delivery Cost
over the years

Figure 3.3

Shipping Challenges and Cost Control

Figure 3.1 shows that Standard Class shipping (8.11% cost-to-sales ratio) has the highest late shipments, while First Class and Same Day shipping (16.83% and 17.38%) provide faster delivery but at higher costs. A balanced approach should prioritise urgent orders for faster shipping while encouraging cost-effective alternatives. Figure 3.2 highlights inefficiencies, with 40.59% of shipments delayed, exceeding the 38.63% on-time rate, while 22.60% arrive early, indicating dispatch issues. Reducing delays is crucial for customer retention. Figure 3.3 shows shipping costs rising from \$34,322 (Q4 2020) to \$159,501 (Q4 2023), driven by fuel prices, supply chain issues, or higher order volumes. Optimising delivery routes and renegotiating contracts is essential for cost control.



Figure 3.4

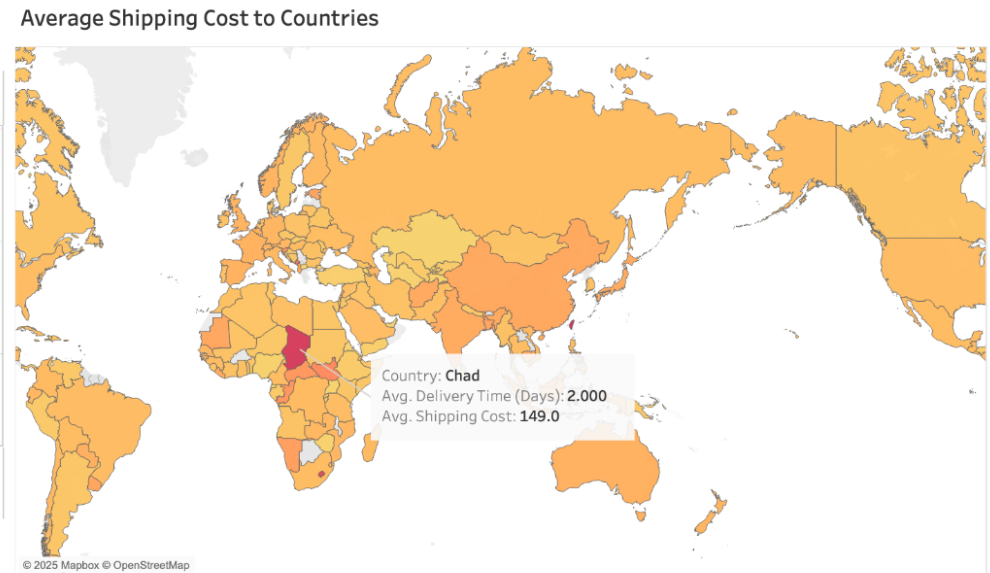


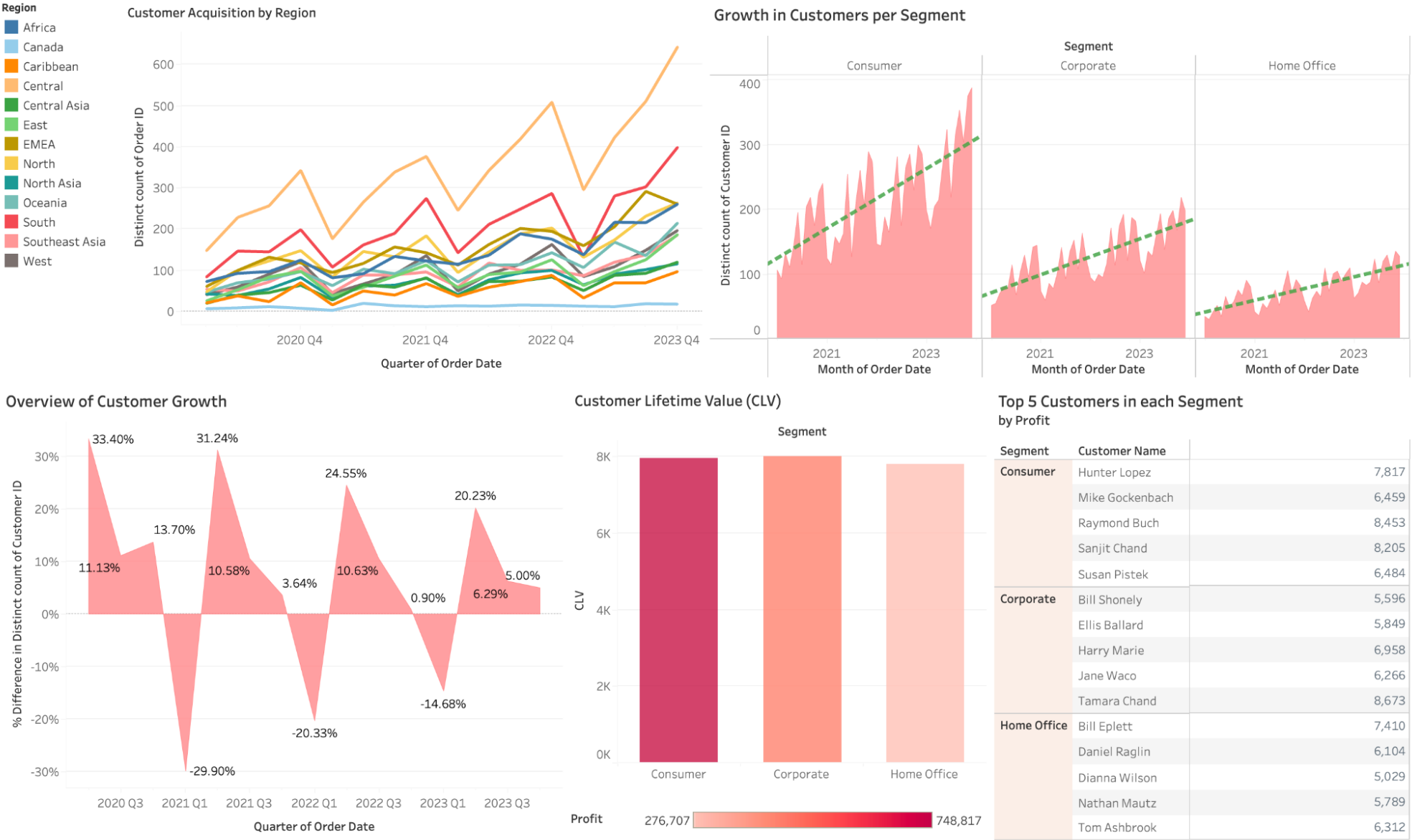
Figure 3.5

Shipping Cost Analysis and Regional Variations

Figure 3.4 reveals that **Standard Class shipping**, while generating the highest revenue, also incurs substantial expenses, eroding profitability. **Same Day and First Class shipping** have disproportionately high costs relative to sales. To mitigate inefficiencies, businesses should **limit costly shipping for low-margin products** and incentivise customers to choose lower-cost options through **discounts and promotions**. Additionally, profit adjusted for shipping costs shows **high expenses correlate with lower profits**, with expedited shipping often leading to minimal or negative margins. Implementing **minimum order values for free or expedited shipping** can help ensure logistics costs do not cut into profitability.

Figure 3.5 highlights regional disparities, with Chad having the highest shipping cost at \$149.90 per order, likely due to **low order volume and lack of economies of scale**. To improve efficiency, businesses should **consider regional distribution centres, bulk shipments, or localised fulfilment strategies** to reduce shipping overheads.

5. CUSTOMER ACQUISITION & RETENTION



Dashboard 4: Customer Acquisition & Retention Dashboard

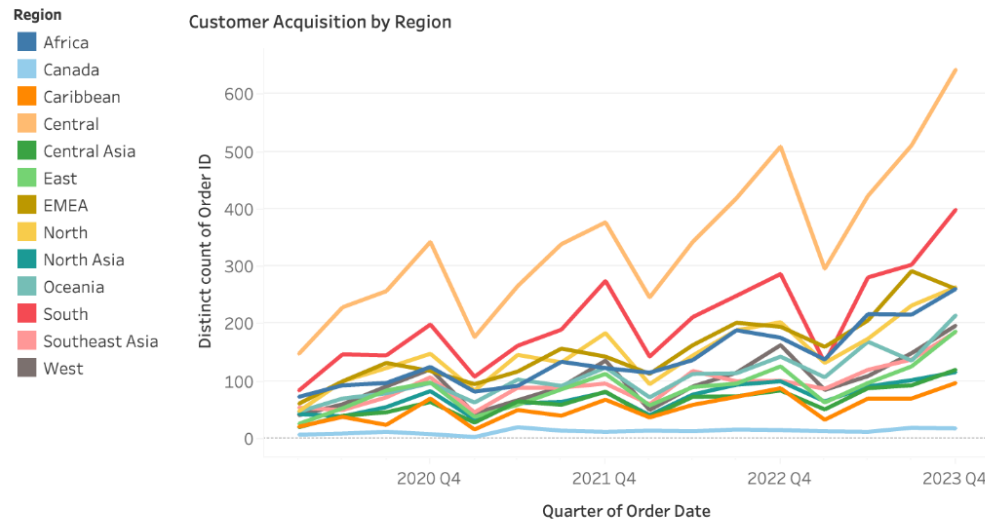


Figure 4.1

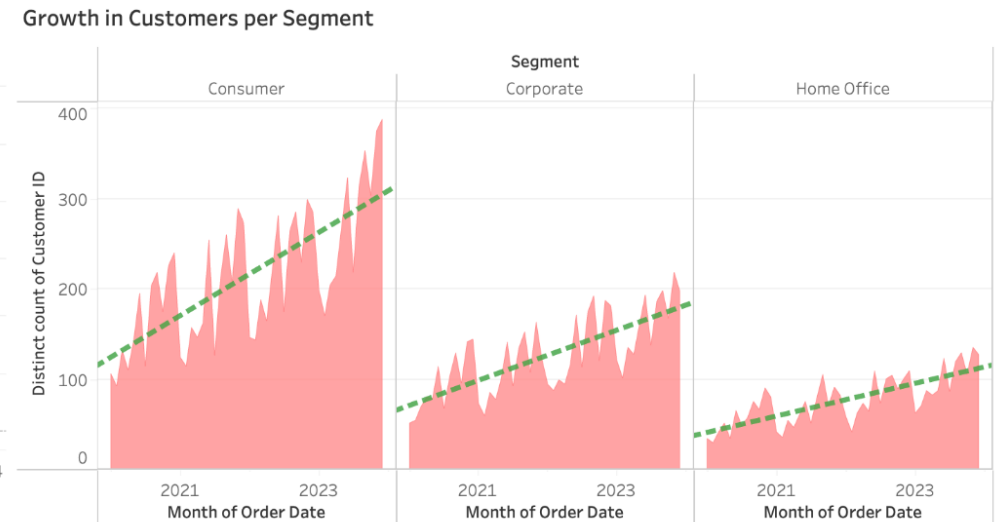


Figure 4.2

Customer Acquisition Trends and Growth Opportunities

Figure 4.1 reveals a general upward trend in customer acquisition across all regions, though Canada shows a relatively slow and horizontal increase, indicating potential for targeted strategies to boost growth. Similarly, Figure 4.2 shows a steady rise in customers across all segments from 2020 to 2023. However, the Home Office segment exhibits a flatter upward trend, suggesting slower growth compared to other segments. These insights highlight opportunities for region- and segment-specific initiatives to enhance customer acquisition.

Overview of Customer Growth

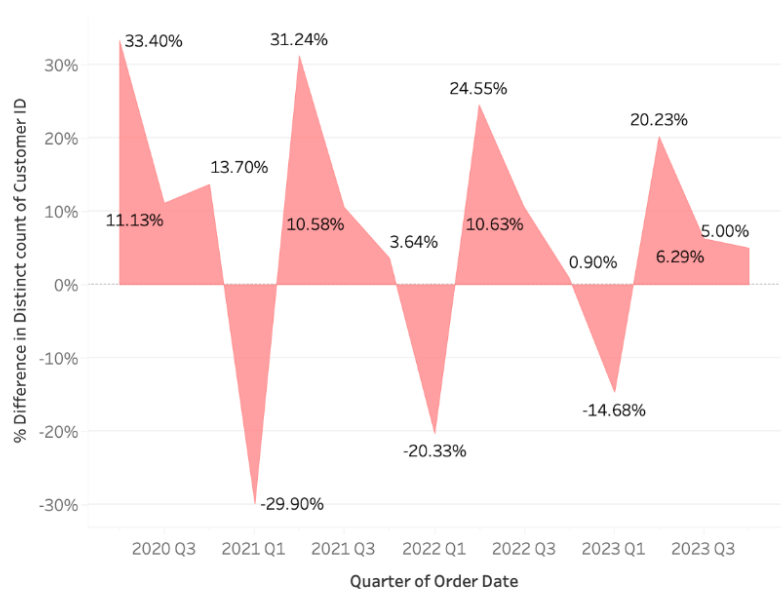


Figure 4.3

Customer Lifetime Value (CLV)

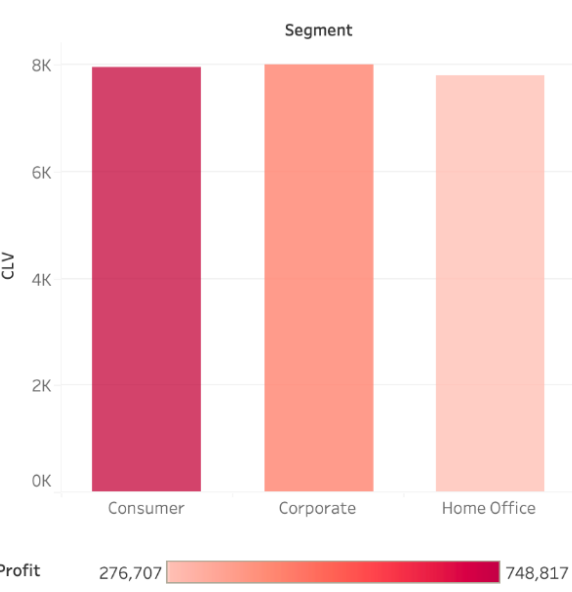


Figure 4.4

Top 5 Customers in each Segment by Profit

Segment	Customer Name	Profit
Consumer	Hunter Lopez	7,817
	Mike Gockenbach	6,459
	Raymond Buch	8,453
	Sanjit Chand	8,205
	Susan Pistek	6,484
Corporate	Bill Shonely	5,596
	Ellis Ballard	5,849
	Harry Marie	6,958
	Jane Waco	6,266
	Tamara Chand	8,673
Home Office	Bill Eplett	7,410
	Daniel Raglin	6,104
	Dianna Wilson	5,029
	Nathan Mautz	5,789
	Tom Ashbrook	6,312

Figure 4.5

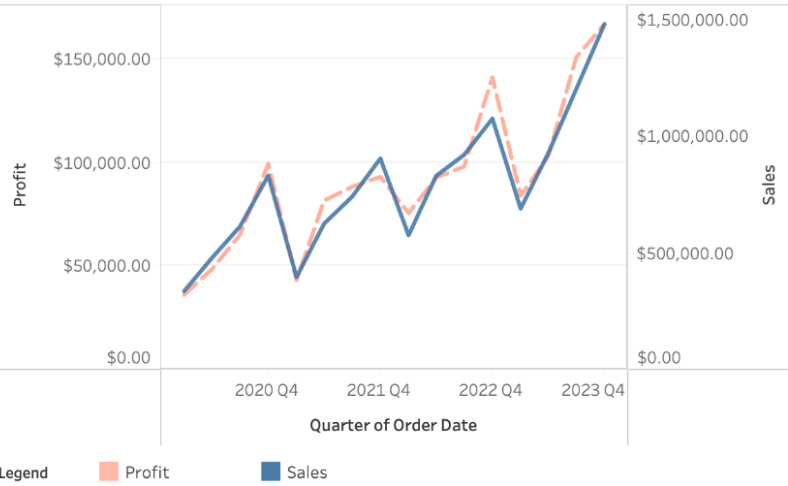
Customer Growth Challenges and Retention Strategies

Figure 4.3 shows a -29.90% decline in customer growth in Q1 2021, likely due to COVID-19, causing economic uncertainty, supply chain disruptions, and reduced consumer spending, especially in retail and office-related sectors. Figure 4.4 highlights the Consumer segment's high Customer Lifetime Value (CLV), reinforcing the need for strong retention strategies.

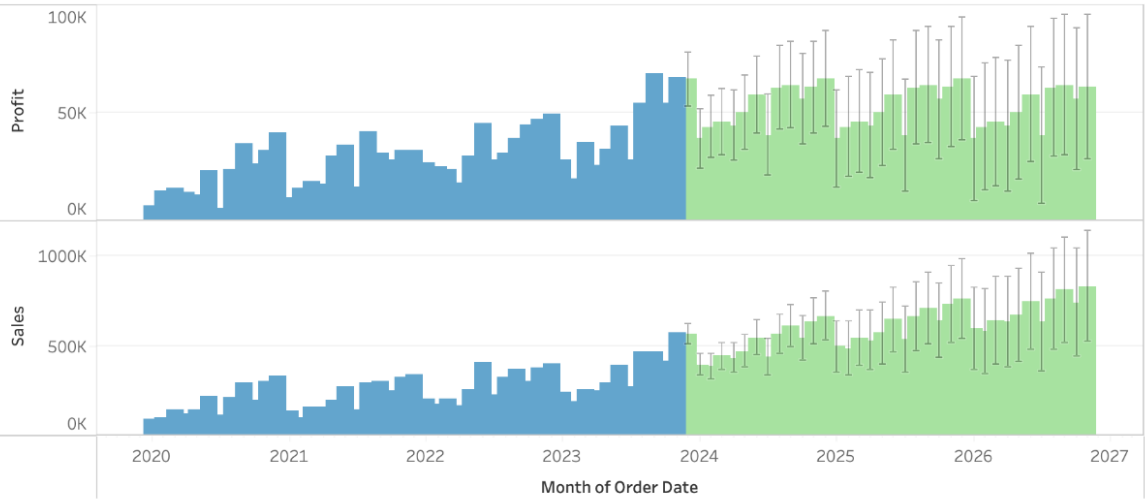
Figure 4.5 identifies the top five most profitable customers per segment, presenting opportunities for targeted retention efforts such as personalised offers, exclusive rewards, and loyalty programs to sustain long-term relationships and maximise profitability.

6. BUSINESS OUTLOOK & FORECAST

Annual Sales & Profit



Sales & Profit Forecast for next 3 years

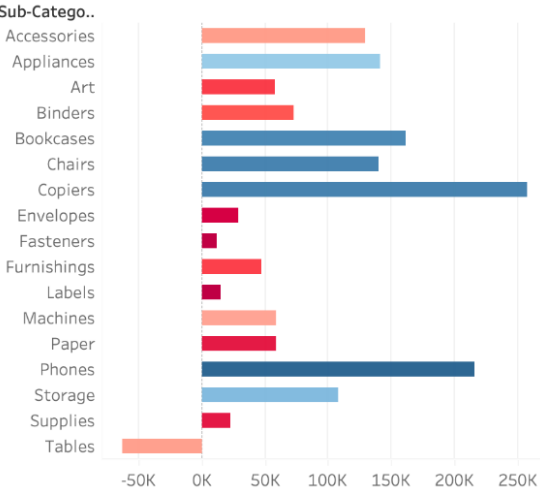


Profit Margin by Region

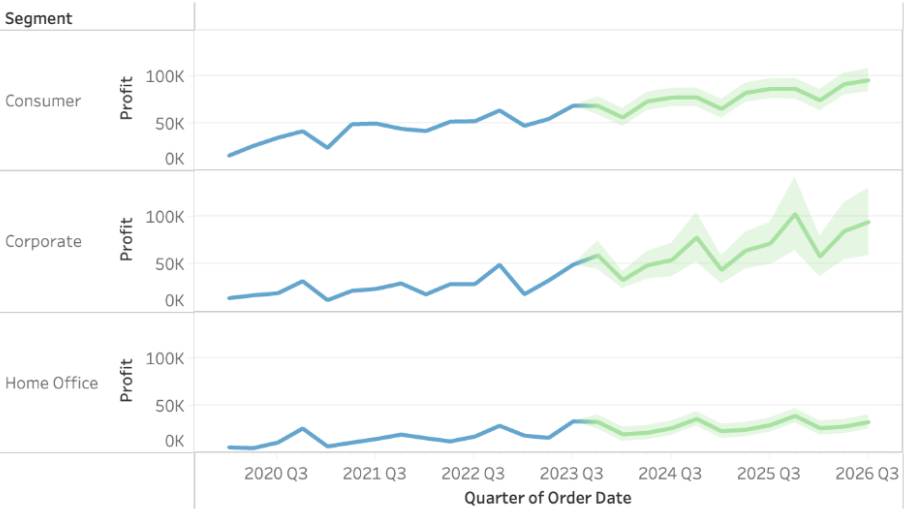
Region	Order Date			
	2020	2021	2022	2023
Africa	8.60%	8.24%	11.54%	13.93%
Canada	21.24%	30.37%	26.77%	25.88%
Caribbean	7.64%	13.48%	9.27%	11.86%
Central	10.18%	10.60%	12.83%	10.31%
Central Asia	18.54%	16.61%	16.91%	18.26%
East	13.26%	13.49%	11.19%	15.57%
EMEA	3.87%	3.29%	5.29%	7.46%
North	16.84%	17.04%	16.63%	13.17%
North Asia	18.26%	18.39%	20.80%	19.94%
Oceania	12.50%	11.91%	11.84%	8.68%
South	6.28%	9.30%	9.03%	9.46%
Southeast Asia	2.16%	1.47%	1.32%	3.12%
West	13.57%	14.58%	12.83%	17.53%

Profit Marg.. 1.32% 30.37%

Product Profitability



Segment Profitability in 3 years



Annual Sales & Profit



Figure 5.1

Sales & Profit Forecast for next 3 years

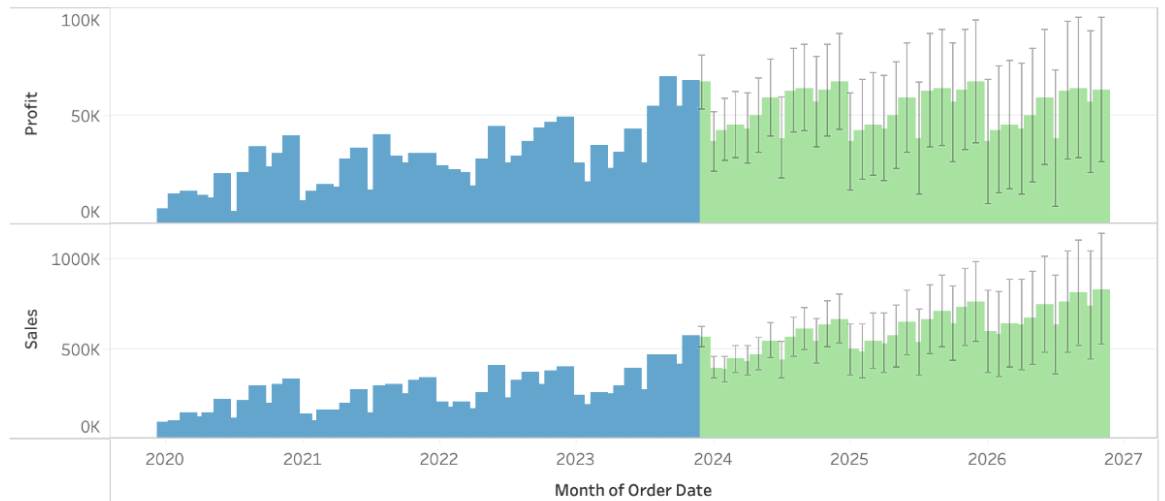


Figure 5.2

Sales and Profit Trends with Future Forecast

Figure 5.1 shows a steady rise in annual sales and profitability from 2020 to 2023, with both metrics following a similar trajectory. While seasonal fluctuations are evident, the overall trend remains positive.

Figure 5.2 presents a forecast for the next three years, indicating continued growth. However, widening error bars suggest potential economic volatility or shifts in consumer demand. To sustain momentum, businesses should prioritise high-margin products and closely monitor market conditions to adapt to emerging challenges and opportunities.

Profit Margin by Region

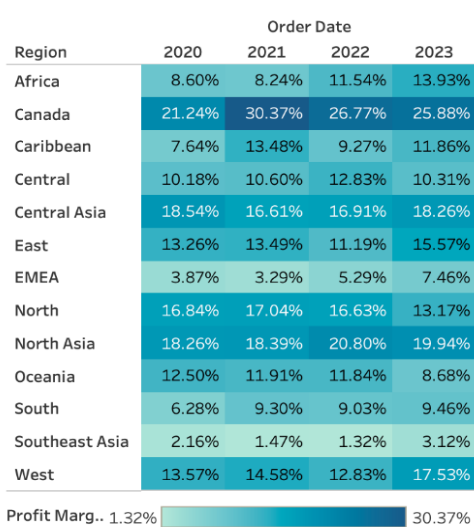


Figure 5.3

Product Profitability

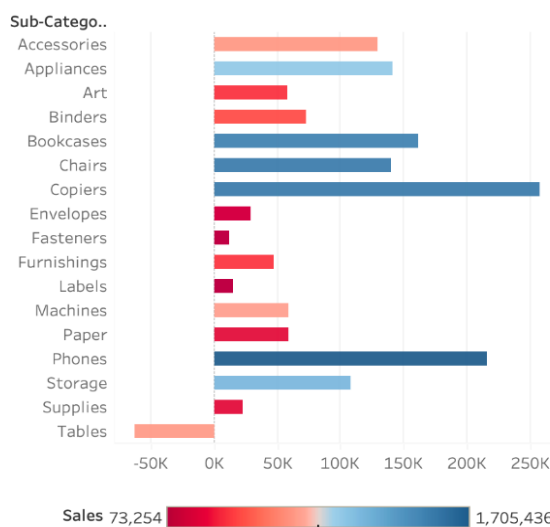


Figure 5.4

Segment Profitability in 3 years

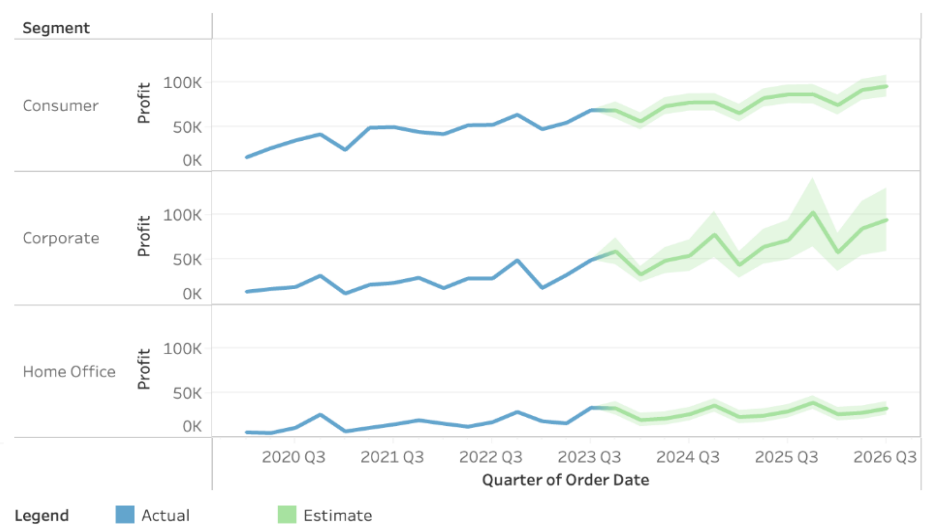


Figure 5.5

Regional and Product Profitability Insights

Figure 5.3 highlights regional profit margin variations, with Canada leading at 30.37% in 2021, though declining to 25.88% in 2023. Central and North America remain strong, while Africa and Southeast Asia struggle with lower margins, and the Western region's drop from 13.57% to 7.53% signals inefficiencies. Businesses should focus on high-margin markets and optimise weaker regions' pricing and operations.

Figure 5.4 identifies Copiers and Phones as top profit generators, whereas Tables, Machines, and Bookcases incur losses, likely due to pricing, manufacturing, or demand issues. Accessories and Binders show seasonal fluctuations, necessitating strategic pricing and portfolio adjustments.

Figure 5.5 reveals Consumer as the most profitable segment, while Corporate has growth potential and Home Office is improving. Expanding Corporate and Home Office segments through targeted promotions and incentives can drive further profitability.

7. RECOMMENDATIONS

Optimise Pricing & Discount Strategies

The dashboards indicate significant profit margin disparities across product categories and regions, suggesting a need for refined pricing strategies. Low-margin categories, such as tables and machines, generate losses despite steady sales, highlighting issues with excessive discounting or high production costs. Implementing dynamic pricing models will allow real-time price adjustments based on regional demand, competitor pricing, and seasonal trends, ensuring optimal profitability. Additionally, the Profit Margin by Region analysis reveals that some markets, such as Canada and North America, maintain high margins, while others struggle. Targeted discounts in high-performing regions rather than across all markets will enhance revenue retention. A tiered discounting approach, offering strategic promotions to high-value customers while minimising broad markdowns, will further balance sales volume and profitability.

Improve Logistics & Shipping Efficiency

The Shipping Analysis Dashboard reveals that express shipping, though used for a small fraction of orders, contributes disproportionately to total logistics costs. Late shipments have also nearly doubled from 2020 to 2023, impacting customer satisfaction and operational efficiency. To mitigate these inefficiencies, shifting non-urgent orders to cost-effective standard shipping can significantly reduce overhead costs without sacrificing reliability. The Delivery Cost Trend graph highlights a steady rise in logistics expenses, necessitating bulk rate negotiations with carriers and shipment consolidation strategies. Additionally, Geographic Shipping Cost Variance data suggests that regions with low order volumes, such as Chad, incur excessive shipping costs. Introducing regional fulfilment centres or bundling shipments in these areas will help optimise expenses. AI-driven route planning and warehouse placement improvements will further streamline supply chain operations, ensuring cost reduction, improved delivery performance, and enhanced customer satisfaction.

8. CONCLUSION

By implementing these strategic recommendations, ShopWave can enhance profitability, operational efficiency, and customer satisfaction. A data-driven approach to pricing and logistics will reduce unnecessary costs and improve market positioning. With these optimisations, ShopWave can drive sustainable growth and maintain a competitive advantage in the e-commerce industry.